

CH 1 — Abstract Vector Spaces

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1.1 Vector Spaces

Ideas: $\mathbb{R}^n$ is useful. There are other objects that “behave” and “look” like $\mathbb{R}^n$

Examples:

1. $V = P_b(\mathbb{F}) = \{a_0 + a_1x + a_2x^2 + \dots + a_nx^n : a_i \in \mathbb{F}\}$

Polynomial degree $\leq n$:

Additive property

$$a_0 + a_1x + \dots + a_nx^n + b_0 + b_1x + \dots + b_nx^n = (a_0 + b_0) + (a_1 + b_1)x + \dots + (a_n + b_n)x^n$$

Scalar Multiplicative property

$$c(a_0 + a_1x + \dots + a_nx^n) = ca_0 + ca_1x + \dots + ca_nx^n$$

2. $V = M_{m \times n}(\mathbb{F}) = \left\{ \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \dots & \dots & \dots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} : a_{ij} \in \mathbb{F} \right\}$

Same properties for matrices like the above

Observation:

$\mathbb{R}^4$	$P_3(\mathbb{R})$	$M_{2 \times 2}(\mathbb{R})$
$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$	$a + bx + cx^2 + dx^3$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Notice that the above not only “look alike”, they share similar operations (addition and scalar multiplication)

This is isomorphism

3. $V = F(\mathbb{R}, \mathbb{R}) = \{\text{functions}, f : \mathbb{R} \rightarrow \mathbb{R}\}$

We can add and scalar multiply

Question: Does $F(\mathbb{R}, \mathbb{R})$ is isomorphic to $\mathbb{R}^n$

ANS: No, the reason will be learned later in the course. The current explanation is that $F(\mathbb{R}, \mathbb{R})$ is an infinity-dimensioned vector space.

Definition – Vector Space

A vector space over $\mathbb{F}$ is a set V together with two operations $\begin{cases} \text{addition}(+) \\ \text{scalar multiplication}(\cdot) \end{cases}$ s.t.

$$\forall \vec{x}, \vec{y}, \vec{x} + \vec{y} \subseteq V$$

$$\forall c \in \mathbb{F}, c\vec{x} \subseteq V$$

Furthermore, the following conditions must hold for $\forall \vec{x}, \vec{y}, \vec{z} \in V, c, t \in \mathbb{F}$

1. $\vec{x} + \vec{y} = \vec{y} + \vec{x}$
2. $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$
3. $c(\vec{x} + \vec{y}) = c\vec{x} + c\vec{y}$
4. $c(t\vec{x}) = ct(\vec{x})$
5. $(s + t) \cdot \vec{x} = s \cdot \vec{x} + t \cdot \vec{x}$
6. $1 \cdot \vec{x} = \vec{x}$
7. $\exists \vec{0} \in V$ s.t. $\vec{x} + \vec{0} = \vec{x}$
8. $\forall \vec{x} \in V, \exists -\vec{x} \in V$ s.t. $\vec{x} + (-\vec{x}) = \vec{0}$

Warning – Zero Vector and Additive Inverse

Zero vector $\vec{0}$ is not the vector with every entry being 0. It is subject to change with its corresponding vector space

Additive Inverse $-\vec{x}$ are not the vector with every entry being the negation of its origin entry. Refer to the Axiom to determine the corresponding additive inverse

Both $\vec{0}$ and $-\vec{x}$ are **unique** in their corresponding vector space

Note:

1. The elements in V are called vectors
2. We omit the “.” in $s \cdot \vec{x}$ to $s\vec{x}$
3. We write $\vec{x} - \vec{y}$ instead of $\vec{x} + (-\vec{y})$
4. $\vec{0}$ is called the **zero vector**
5. $-\vec{x}$ is called the **additive inverse** of $\vec{x}$

Examples:

$$1. V = \mathbb{F}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_i \in \mathbb{F} \right\}, V = M_{m \times n}(\mathbb{F}) = \left\{ \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} : a_{ij} \in \mathbb{F} \right\},$$

and $V = P_n(\mathbb{F}) = \{a_0 + a_1x + \dots + a_nx^n : a_i \in \mathbb{F}\}$ with addition and scalar multiplication be usual operations.

$$\text{Here, } \vec{0} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix}, 0 + 0x + \dots + 0x^n$$

2. $V = F(\mathbb{R}, \mathbb{R}) = \{\text{functions}, f : \mathbb{R} \rightarrow \mathbb{R}\}$ with addition and scalar multiplication be usual operations.

Here, $\vec{0}$ = the zero function, i.e. $z(x) = 0$

3. $V = \mathbb{R}^2$

Define addition and scalar multiplication by

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + y_2 \\ x_2 + y_1 \end{bmatrix}$$

$$s \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} sx_2 \\ sx_1 \end{bmatrix}$$

This is NOT a vector space.

$$1\vec{x} \neq \vec{x} \implies \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \neq \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$$

$$\vec{x} + \vec{y} \neq \vec{y} + \vec{x}$$

4. $V = \mathbb{R}^2$

Define addition and scalar multiplication by

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 - 1 \\ x_2 + y_2 + 1 \end{bmatrix}$$

$$s \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} sx_1 + s - 1 \\ sx_2 - s + 1 \end{bmatrix}$$

Check axiom 7 and 8 as 1-6 are easy and straightforward

Axiom 7:

Note that in this case $\begin{bmatrix} 0 \\ \dots \\ 0 \end{bmatrix}$ is not the zero vector for this vector space!

We suppose $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$,

$$\text{then } \vec{x} + \vec{0} = \vec{x} \iff \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \iff \begin{bmatrix} x_1 + z_1 - 1 \\ x_2 + z_2 + 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \implies \vec{z} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \in V$$

Axiom 8:

We suppose $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$

$$\text{then } \vec{x} + \vec{z} = \vec{0} \iff \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \iff \begin{bmatrix} x_1 + z_1 - 1 \\ x_2 + z_2 + 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies \vec{z} = \begin{bmatrix} x_1 + 2 \\ x_2 - 2 \end{bmatrix}$$

$$-\vec{x} = \begin{bmatrix} -x_1 - 2 \\ -x_2 + 2 \end{bmatrix}$$

5. Proof uniqueness of $\vec{0}$

Let V be a vector space. Suppose $\vec{0}_1, \vec{0}_2 \in V$ satisfy

$$\vec{0}_1 + \vec{x} = \vec{x}, \vec{x} \in V$$

$$\vec{0}_2 + \vec{x} = \vec{x}, \vec{x} \in V$$

We want $\vec{0}_1 = \vec{0}_2$

Proof

$$\vec{0}_2 + \vec{x} + \vec{0}_1 = \vec{x} \implies \vec{0}_2 + \vec{0}_1 = \vec{0}_1$$

$$\vec{0}_1 + \vec{x} + \vec{0}_2 = \vec{x} \implies \vec{0}_1 + \vec{0}_2 = \vec{0}_2$$

$$\text{Together, } \vec{0}_1 + \vec{0}_2 = \vec{0}_2 + \vec{0}_1 = \vec{0}_1 = \vec{0}_2$$

□

6. Proof $0\vec{x} = \vec{0}, \forall \vec{x} \in V$

$$0\vec{x} = (0 + 0)\vec{x} = 0\vec{x} + 0\vec{x} \implies \vec{0} = 0\vec{x}$$

1.2 Subspaces and Spans

Definition – Subspace

Let V be a vector space over $\mathbb{F}$ and $W \subseteq V$. We say **subspace** of V if W is also a vector space over $\mathbb{F}$ with same addition and scalar multiplication as for V

Info – Subspace Test

Let V be a vector space over $\mathbb{F}$. Let $W \subseteq V$.

Then W is a subspace of V if and only if:

1. $\vec{0} \in W$
2. W is closed under addition (if $\vec{x}, \vec{y} \in W, \vec{x} + \vec{y} \in W$)
3. W is closed under scalar multiplication (if $\vec{x} \in W, c \in \mathbb{F}, c\vec{x} \in W$)

Example:

1. Let $W = \{p(x) \in P_0(\mathbb{R}) : p'(0) = 0\}$

W is a vector space with usual $+$ and $\cdot$ of polynomials

Most of the axioms are obvious for all polynomials.

We need to check axioms that are less intuitive:

1. Is the zero vector in W ?
2. Is W closed under addition?
3. Is W closed under scalar multiplication?

Proof

Let $V \in P_n(\mathbb{R})$

1. Is the zero vector in W ?

$$0 + 0x + \dots + 0x^n = z(\vec{x})$$

Clearly, $z'(0) = 0$ So $z(\vec{x}) \in W$

2. Is W closed under addition?

Let $p(x), q(x) \in W$

$$(p + q)'(0) = p'(0) + q'(0) = 0 \in W$$

3. Is W closed under scalar multiplication?

Let $p(x) \in W, s \in \mathbb{R}$.

$$(sp)'(0) = s(p'(0)) = s0 = 0 \in W$$

□

2. Let V be any vector space.

$W = V$ is a subspace. $W \setminus \{\vec{0}\}$ is a subspace.

3. $V = \mathbb{F}^n$

The origin is a subspace.

The lines pass through origin are subspace.

The planes pass through origin are subspace.

4. $V = P_3(\mathbb{F}), W = \{a + bx + cx^2 + dx^3 : b \geq 0\}$

W is not a subspace: not closed under scalar multiplication.

$$2x \in W, -1(2x) \notin W$$

4. $V = P_n(\mathbb{F}), W = \{p(x) \in V : p''(1) + p'(2) = 2\}$

W is not a subspace since $\vec{0} \notin W$

Note:

- A tip is that **non-homogeneous and equations** are usually **not** a subspace
- **Linear homogeneous** equations are usually a subspace

Definition – Span

Let V be a vector space. Let $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \in V$.

Then the **span** of $\vec{v}_1, \dots, \vec{v}_k$ is the set

$$W = \text{span}\{\vec{v}_1, \dots, \vec{v}_k\} = \{a_1\vec{v}_1 + \dots + a_k\vec{v}_k : a_i \in \mathbb{F}\}$$

Theorem

Let V be a vector space, let $\vec{v}_1, \dots, \vec{v}_k$ then $W = \text{span}\{\vec{v}_1, \dots, \vec{v}_k\}$ is a subspace of V

Note: the example (1) below gives a second proof that $\{A \in M_{2 \times 2} : \text{tr}(A) = 0\}$ is a subspace of $M_{2 \times 2}(\mathbb{F})$

Example:

1. $V = M_{n \times n}(\mathbb{F}), W = \{A \in M_{n \times n} : \text{trace}(A) = 0\}$

Recall that $\text{trace}(A)$ is the sum of the diagonal (i.e. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \text{trace}(A) = a + d$)

Claim W is a subspace of V

Recall that

- $\text{trace}(A + B) = \text{trace}(A) + \text{trace}(B)$
- $\text{trace}(cA) = c \text{ trace}(A), c \in \mathbb{R}$

Proof

$$1. \vec{0} = \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix}, \text{trace}(\vec{0}) = 0 + \dots + 0 = 0 \implies \vec{0} \in W$$

2. Let $A, B \in W, c \in \mathbb{F}$.

$$\text{trace}(cA + B) = \text{trace}(cA) + \text{trace}(B) = c \text{trace}(A) + \text{trace}(B) = 0 \text{ since } A, B \in W$$

Thus $cA + B \in W$, closed by addition and scalar multiplication.

□

Take $W = \{A \in M_{2 \times 2}(\mathbb{F}) : \text{trace}(A) = 0\}$

For an arbitrary $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in W$

$$\iff \text{trace}(A) = 0 \iff a + d = 0 \iff d = -a \iff A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & -a \end{bmatrix} (a, b, c \in \mathbb{F})$$

$$\text{So } W = \left\{ \begin{bmatrix} a & b \\ c & -a \end{bmatrix} : a, b, c \in \mathbb{F} \right\} = \left\{ \begin{bmatrix} a & 0 \\ 0 & -a \end{bmatrix} + \begin{bmatrix} 0 & b \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ c & 0 \end{bmatrix} : a, b, c \in \mathbb{F} \right\}$$

$$\implies W = \text{span} \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$$

We can conclude that most of the time **imposing condition decreases degrees of freedom**

Note: differentiation and equation are **linear** operations

$$2. V = P_2(\mathbb{R}), W = \{p(x) \in V : p'(1) = p(-1)\}$$

For W , an arbitrary $p(x) = a + bx + cx^2 \in W$

$$p'(x) = b + 2cx \implies p'(1) = b + 2c, p(-1) = a - b + c$$

$$\text{Then } p'(1) = p(-1) \iff b + 2c = a - b + c \iff a - 2b + c = 0$$

$$\iff a = 2b + c \iff p(x) = a + bx + cx^2 = (2b + c) + bx + cx^2$$

$$\text{Now } W = \{(2b + c) + bx + cx^2 : b, c \in \mathbb{F}\} = \{b(2 + x) + c(1 + x^2) : b, c \in \mathbb{F}\}$$

$$\text{So } W = \text{span}\{2 + x, 1 + x^2\}$$

Note:

1. Writing a subspace W in the form $W = \text{span}\{\vec{v}_1, \dots, \vec{v}_k\}$ is that it is easier to write down vectors in W

2. Having $W = \text{span}\{\vec{v}_1, \dots, \vec{v}_k\}$ allows us to detect the $\dim W$. Here is a catch, some spanning set are better than others

(i.e. $W = \text{span}\{2 + x, 1 + x^2\}$, instead someone came up with

$W = \text{span}\{2 + x, 1 + x^2, 3 + x + x^2\}$, the latter one, the span is linearly dependent)